Sensor applications

Force Myography for Motion Intention Detection Based on 3D-Printed Piezoelectric Sensors

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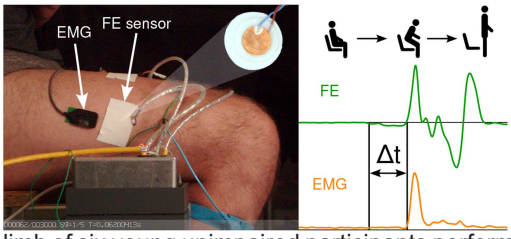
Abstract—Recognizing human movement intention is key to control wearable robotics for assistance, such as in the elderly and rehabilitation. Piezoelectric sensors, in particular highly sensitive ferroelectrets (FEs), qualify for the assessment of subtle muscle movements using force myography (FMG). In contrast to electromyography (EMG), which measures electrical muscle activity, FMG measures the mechanical muscle response to overcome issues in electrode placement and skin preparation. We employ 3D-printed polypropylene FE sensor patches for FMG adjacent to EMG electrodes as a reference at the lower limb of six young unimpaired participants performing sit-to-stand movements. The activity onsets of FMG and EMG are systematically compared at the quadriceps femoris and visually verified using a high-speed camera. The average time difference (35 ± 50 ms) aligns with the known electromechanical muscle delay. Notably, our results demonstrate that FMG can provide an early indicator for the activity onset significantly before the EMG (on average 263 ± 90 ms earlier) in this scenario due to the possibility to measure passive muscle deformation. In addition, the signal-to-noise ratio of the FE signal (53 dB) exceeds that of the EMG (39 dB) during the maximum muscle activity, while the average variability is slightly larger for FE (22%) compared to EMG (17%). In conclusion, our customizable FE sensors prove to be suitable for motion intention detection and promise an advance in wearable assistive devices, such as active prostheses and exoskeletons, to support people in need.



Index Terms—Sensor applications, electromyography (EMG), ferroelectret, force myography (FMG), muscle activity, sit-to-stand, wearable.

I. INTRODUCTION

Muscle weakness due to age or diseases can result in major limitations in everyday life. Assistive systems, such as exoskeletons and prostheses, offer the opportunity to support the user in the intended motion and restore quality of life [1]. Wearable robotics for movement assistance requires sensors for the detection of the user's movement intention that are suitable for everyday use to provide appropriate and synchronized support [2]. One of the frequent movements that requires among the largest forces and causes special difficulties in the elderly is the transition from sitting to standing, the sit-to-stand movement [3]. It can be divided into three partially overlapping movement phases based on the kinematics [3], [4]. The trunk's flexion initiates the movement and shifts the center of mass anterior, followed by the seat-off coinciding with a knee extension, and concluding with a trunk extension until reaching the upright posture. Detecting the intention to stand up as early as possible is crucial to provide support throughout the entire movement duration, particularly during the early phase before seat-off when the quadriceps muscle exerts maximal extension forces at the knee [3].

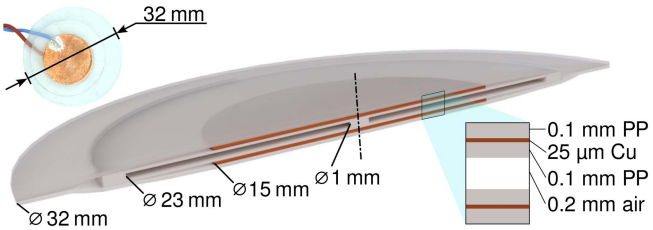


Fig. 1. Piezoelectric sensors for motion intention detection at muscle level are 3D-printed highly sensitive ferroelectrets with well-defined charged air cavities. They consist of flexible structured polypropylen (PP) layers, self-adhesive copper electrodes, and a PP insulation cover.

Here, we consider motion intention detection in terms of a peripheral indicator solely based on muscle activity, distinct from the cognitive intent to perform an action [5]. Motions are triggered by electrical stimulation of muscles through action potentials, measurable by electromyography (EMG) [6]. After an electromechanical delay of

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about 30–100 ms [7], the muscle contraction leads to a deformation, detectable with optical, acoustic, and force sensors [8], [9].

Currently, EMG is a widely used method for muscle activity measurements in clinical and research settings [10]. It exhibits an inherent time advantage due to the electrical signal preceding the mechanical deformation. However, it is prone to mechanical stress, sweat, and electromagnetic interference, resulting in reduced signal quality or even signal loss [10], [11]. Firm skin contact is critical for long-term measurements, limiting its applicability in everyday use.

Mechanical sensor technologies promise an attachment to the human skin with reduced preparation requirements and higher resilience. Especially, force and pressure sensors utilized in force myography (FMG) qualify for wearable devices due to their small dimensions, remote usability, and immunity to visual obstruction. FMG measures the mechanical muscle response and is able to assess passive muscle movements without muscle contraction, which are not detectable by EMG. Commonly used FMG sensors include force-sensing resistors [12], [13], [14], which have known drawbacks in accuracy and repeatability due to creep behavior [15]. Furthermore, FMG is predominantly employed with elastic straps or bandages [12], [13], [14], [16], causing the muscles around the entire limb circumference to act on each sensor at least partially. Other sensor technologies for FMG include capacitive [16], optical [17], and piezoelectric [18], [19] sensors. In general, lower limb FMG applications are underrepresented in the literature [9], as most FMG research concerns forearm movements for activities such as hand gesture recognition [20], [21], [22].

Piezoelectric sensors qualify for measuring fast movements due to their dynamic working principle. In particular, ferroelectret (FE) sensors, which are porous polymer structures with charged air cavities, exhibit highest sensitivities [23], [24], [25] and require considerable less pressure for activation than other sensors in FMG. Recently, polymers such as polyalactide (PLA) and polypropylene (PP) are utilized in FE sensors, enabling fast application-specific customization [20], [21], [26], [27]. The flexibility of these materials facilitates their application as wearable sensor patches, eliminating the need for elastic straps or bandages, to detect subtle mechanical muscle contractions.

Consequently, this study investigates FE sensors for detecting motion intention as early as possible to substitute or supplement the widely used EMG. We evaluate the highly sensitive sensor technology in a sit-to-stand movement, while obtaining EMG and visual data as a reference to interpret the signals. Two movement speeds are assessed, comprising an average motion with nominal speed and a slower motion that resembles the reduced speed in the elderly.

II. METHODS

A. Sensor Fabrication and Measurement Setup

The FE sensors are 3D-printed from PP using default print settings and are based on two 0.1 mm-thick disks (diameter 25 mm). One of the disks includes an additional structured layer (0.2 mm thick), consisting of a 23 mm-wide circumference and a center point (diameter 1 mm), to create well-defined cavities between the two disks, which are thermally bonded (Fig. 1). Self-adhesive copper foil on both sides (diameter 15 mm) forms the electrodes. The structure is electrically treated by contact charging, utilizing a DC voltage of 5 kV over a period of 120 s. The high voltage applied causes Paschen breakdown in the air cavities, resulting in quasi-permanently trapped charges on the material surfaces inside the cavities [23]. Finally, the FE sensors are insulated with two 0.1 mm-thick PP covers (diameter 32 mm). Shielded twisted pair cables connect the FE sensors for FMG measurements to our custom electronics designed for impedance matching and signal conditioning to yield an output voltage proportional to the sensor deformation velocity [26]. Electrical muscle activity is acquired as a reference using a commercially available EMG system (Bagnoli, Delsys, Natick, MA, USA), with the main amplifier's gain set to 1000. All conditioned FMG and EMG signals are synchronously digitized at a sampling rate of 1 kHz using a data acquisition (DAQ) device (USB-6212, National Instruments, Austin, TX, USA) and transmitted to a PC for further processing. Additionally, a high-speed camera (Chronos 1.4, Kron Technologies, Burnaby, BC, Canada), which is synchronized using a trigger signal, captures the sit-to-stand movement at a frame rate of 1 kHz.

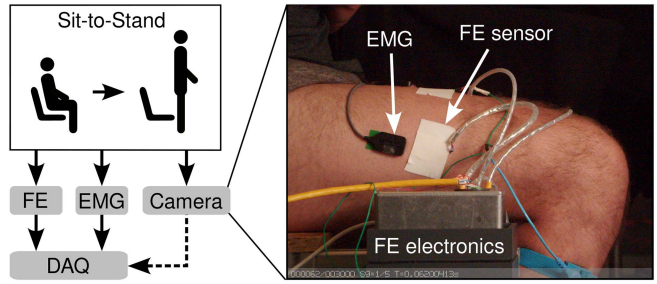

Fig. 2. Experimental setup for data acquisition (DAQ) with three ferroelectret (FE) sensors and electromyography (EMG) electrodes at the vastus medialis, rectus femoris, and vastus lateralis during the sit-to-stand movement. Synchronized EMG data and recordings from a high-speed camera serve as a reference to interpret the FE signals from participants performing movements at nominal and slow speed.

B. Experimental Protocol

Six physically unimpaired participants (two female, four male, 24.5 ± 2.4 years, 180 ± 7 cm, 72 ± 6 kg) are part of this study. The study received a positive vote from the Ethics Committee of Technische Universität Darmstadt and the participants provided written informed consent. The experimental setup consists of a seat with integrated camera mount, focusing on the right thigh in a sagittal view (Fig. 2). FE sensors and EMG electrodes are placed in pairs on the muscles rectus femoris (RF), vastus medialis (VM), and vastus lateralis (VL), following the SENIAM recommendations. This procedure includes shaving and cleaning the relevant skin areas with medical alcohol, which is required for EMG only. The FE sensors are attached using non-elastic medical tape (Leukotape classic, BSN medical, Hamburg, Germany) to mitigate the influence from other muscles. The EMG electrodes are affixed using the Bagnoli adhesive, and the EMG reference electrode is placed on the right ankle.

Each participant performs 20 sit-to-stand movements at nominal speed (1 s) and 10 at slow speed (2 s) [3]. Before the experiment, the movement speeds are demonstrated and participants perform several test movements to internalize the desired speeds and to familiarize with the setup. Instructions to stand up are given without prior notice to ensure natural movement patterns.

C. Data Analyses

All data analyses are performed offline using MATLAB (2022b, MathWorks, Natick, MA, USA). The EMG signals are processed using high-pass filtering (20 Hz), rectification, and subsequent low-pass filtering (10 Hz), which is a standard procedure to obtain a measure of muscle activity [6]. The FE signals need no filtering, but we apply a 10 Hz low-pass filter for EMG comparability. All filters are designed as zero-phase Butterworth types of fourth order.

We determine the times of the muscle activity onset for EMG and FMG, respectively, and an early indicator as third characteristic, which

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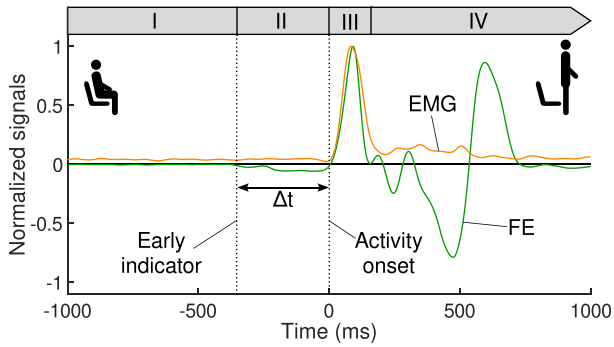

Fig. 3. Representative single sit-to-stand movement measured with a ferroelectret (FE) sensor and electromyography (EMG) at the rectus femoris. Signals are normalized to their maxima, respectively, for visual comparability. The participant starts the sit-to-stand movement from rest (I) through a trunk flexion, shifting the center of mass anterior (II), followed by knee extension and seat-off (III), and concluding with the trunk extension (IV) until upright posture is reached.

exists only in the FMG signal. The activity onset for the EMG signal is the first deviation from the initial resting state at the beginning of a sit-to-stand movement. The activity onset is extracted using a threshold method with a sliding window approach (window length of 50 ms). It is defined by the window center when the moving mean first exceeds the baseline mean by five times the baseline standard deviation (SD). The baseline mean and SD are determined using a 400 ms interval from the resting state. The selected parameters are determined empirically for automated feature extraction.

The activity onset for the FE signal corresponds to the sensor deformation due to muscle contraction. We define it as the FE sensor's zero-crossing, which represents a direction reversal of the sensor deformation, directly before the first major signal peak. In addition, FMG is able to capture passive muscle movements before the muscle force generation at the quadriceps, providing an early indicator for an imminent sit-to-stand movement. This third characteristic is the earliest time of activity in the FE signal, which is determined using the same threshold method as for the EMG activity onset.

The automatically extracted features, i.e., activity onset for both modalities and early indicator for FMG, are examined for algorithmic discrepancies and corrected manually if required. Furthermore, the plausibility of the extracted features is verified by visually inspecting the camera recordings. We calculate the time differences between both FMG features and the EMG activity onset of the nominal speed movements, respectively, to evaluate our sensor in the sit-to-stand scenario. The signal variability is analyzed in the relevant region, i.e., from early indicator to the end of the main activity, as ratio of the SD to the mean peak output. SNR is defined as the ratio of the RMS value in the respective regions compared to the RMS value in the rest phase.

III. RESULTS AND DISCUSSION

The sit-to-stand data reveals distinct peaks across all participants for both FE and EMG signals. The screening for algorithmic discrepancies in the automatically extracted features results in slight corrections for EMG (19% of the cases), FE activity onset (21%), and FE early indicator (3%). The signal patterns clearly reveal the initial rest phase (I) during which the participant is sitting and the subsequent motion phases (II–IV) of the sit-to-stand movement (Fig. 3). The FE sensor exhibits an early indicator in the form of a decreasing signal (II), whereas the EMG indicates no electrical muscle activity at this time. Despite slight user-specific variations in amplitude and

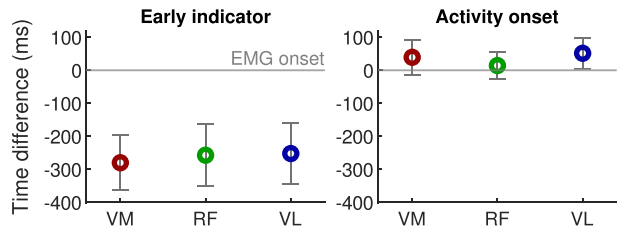

Fig. 4. Time difference between onset features of ferroelectret (FE) sensors and electromyography (EMG) reference at vastus medialis (VM), rectus femoris (RF), and vastus lateralis (VL) with mean and standard deviation across six participants, each performing 20 nominal-speed sit-to-stand movements. Negative values denote an earlier detection in the FE sensor data.

patterns, the early indicator in the FE sensor data is consistent across all participants and speeds. The analysis of the camera recordings reveals a passive muscle deformation at the thigh when the participants flex their trunk (II) to shift the center of mass anterior in preparation of the knee extension, causing the early indicator in the FE signal. The major peak in the EMG signal (III) describes the largest muscle activity in the quadriceps that is necessary for the knee extension and eventually leads to the seat-off. The activity onset indicates the beginning of the quadriceps muscle contraction, which is visible in EMG and FMG. These synchronous features in both signals are uniformly present across all trials. The subsequent trunk extension (IV) partially overlaps with the knee extension. The EMG measures fluctuating muscle activity with a larger baseline than in rest, whereas the FE signal oscillates with relatively large amplitude. This signal shape varies across participants and may originate from uncontrolled tissue movement occurring during the muscle relaxation and stretching [12].

The raw data from the EMG provides on average an SNR of 39 ± 21 dB over the main activity phase (III), whereas the FE sensor demonstrates superior noise resilience with 53 ± 21 dB SNR. Despite the smaller FE sensor amplitude in the early indication phase (II), the signal offers 27 ± 14 dB SNR for a reliable detection. The reliability of the motion intention detection may be further improved using sensor fusion of FMG and EMG data [22].

Compared to the EMG, the FMG provides a significantly earlier detection (paired samples t–tests, $p < 0.001$) of an imminent sit-to-stand movement. The FE sensor's early indicator occurs on average 263 ± 90 ms prior to the EMG activity onset (VM 280 ms, RF 257 ms, VL 252 ms; Fig. 4). A movement can be detected the earliest when the trunk starts to move, on average 508 ms (at around 25% of the motion) before the first lower limb muscles actuate [3]. This literature value confirms the plausibility of our results. In contrast, the activity onset occurs first with EMG-based measurements. In the FE signals, this feature is detected slightly later (average delay 35 ± 50 ms; paired samples t–tests, $p < 0.001$) across all muscles (VM 39 ms, RF 14 ms, VL 51 ms). This aligns with the underlying muscular activation mechanism that causes an electromechanical delay of up to 100 ms [7].

Across all participants and measured muscles, the FE sensor signals at nominal speed exhibit sharp edges suitable for the activity detection and an average variability of 12% and 22% during early indicator and main activity, respectively (Fig. 5). The EMG variability during main activity is slightly lower at 17%. The slow sit-to-stand movement demonstrates consistency with the nominal speed, apart from smaller activity maxima and an elongated movement duration. In particular, the last phase before upright posture exhibits larger variability due to the different durations. However, this part of the signal is less relevant for the application of motion intention detection. Consequently, the slow motion, which resembles the sit-to-stand duration of an elderly person [3], reveals the same features and is detectable using the procedure we presented.

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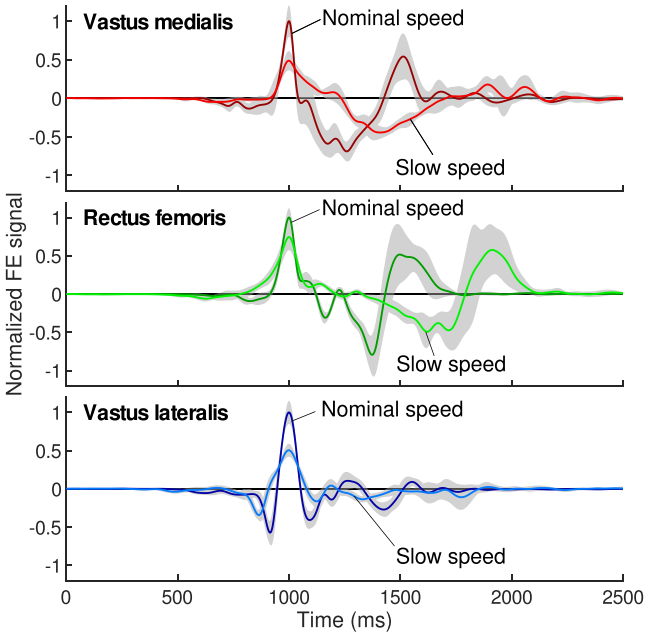

Fig. 5. Mean of the ferroelectret (FE) sensor signals from 20 nominal (1 s) and 10 slow (2 s) sit-to-stand movements of one participant with standard deviaiton (shaded). Signals are normalized to the nominal speed's maximum and synchronized to the major peaks. All muscles exhibit larger peak output at nominal compared to slow speed.

IV. CONCLUSION

FMG using our FE sensors proves to be effective for motion intention detection in the sit-to-stand scenario across multiple participants and speeds. The high sensitivity of the FE sensors and their capability to detect passive muscle movements allow for the detection of characteristic signal features that serve as an early indicator for sit-to-stand movements. On average, the motion intention is detected 263 ms earlier compared to the detection using EMG signals. The major activity peaks of the FE and the EMG signals demonstrate high similarity, making FE sensors a suitable supplement for sensor fusion or substitution of EMG. Individual sensor patches instead of the common sensor straps maintain the possibility to extract distinctive activities of neighboring muscles. Sit-to-stand movements at nominal and slow speed result in comparable signal patterns, confirming the feasibility of FMG at reduced movement speeds as found in the elderly. Future work will examine motion intention detection in other application scenarios to facilitate usage in daily life. In addition, we will investigate the time advantage by the early indicator in an exoskeleton-assisted sit-to-stand scenario. Altogether, motion intention detection based on FE sensors holds the potential to advance the development of assistance systems, thereby providing crucial support to people in need.

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